DSP Equations and Proofs

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28th June 2014

Abstract

These are summary notes for ELEC3104. This is not everything that you need to know. The crucial non-trivial equations which must be rote learned have been boxed. More notes for ELEC3104 and other subjects can be found at www.elsoc.net/notes.php.

Equations and Notes

Quantization and SQNR

Quantization Error

Quantization errors are uniformly distributed between $e = \pm\frac{\Delta V}{2}$, with mean 0.
Where ΔV is the value of the least significant bit.
Quantization noise power is the variance of this.

$$\begin{aligned}\sigma_e^2 &= \int_{-\infty}^{\infty} P(e)\, e^2 de \\ &= \int_{e=-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} \frac{1}{\Delta V} \cdot e^2 de \\ &= \frac{1}{\Delta V}\left[\frac{e^3}{3}\right]_{e=-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}}\end{aligned}$$

Which comes to

$$\boxed{\sigma_e^2 = \frac{(\Delta V)^2}{12}} \tag{1}$$

Forms

Direct Form I

Feed forward before feed back.

Direct Form II

Feed back before feed forward, with uncombined delays.

Canonical

Feed back before feed forward, then combine the delays.

The definition of Canonical is the form which contains the minimum number of delay elements necessary to realise the associated frequency response.

Parallel

Use partial fractions to obtain 1st and 2nd order[1] blocks (each in Canonical form), then add the blocks geometrically in parallel.

Cascade

Factorize to obtain 1st and 2nd order blocks (each in Canonical form), then add them geometrically in series (transfer function multiplication).

Stability

A linear time invariant (LTI) system is stable iff the sum of the absolute value of the impulse response is finite.

$$S = \Sigma_{k=-\infty}^{\infty} |h[k]| < \infty$$

Where $h[k]$ is the impulse response of the system.

Triangle of Stability

For a second order system:

$$H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{z^2}{z^2 + b_1 z + b_2} \tag{2}$$

For stability the poles must be inside the unit circle. You should be able to do the maths[2] by finding the roots of equation 2. The final inequalities are:

[1] The numerator and denominator can both be up to 2nd order. If possible break them down into 1st order blocks.

[2] The full proof is covered on page 11.

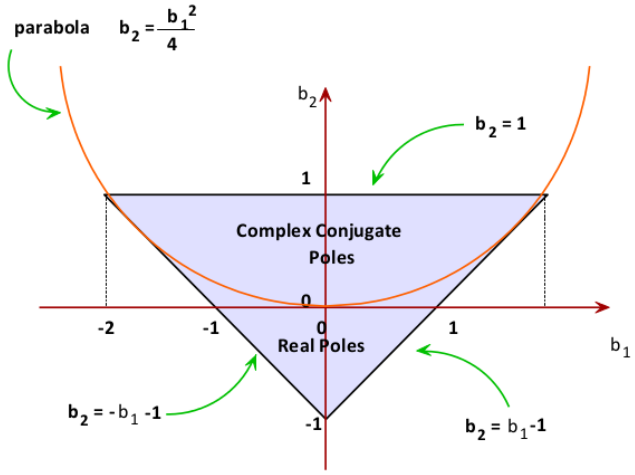


Figure 1: Triangle of stability

$$\boxed{b_2 < 1} \tag{3}$$

$$\boxed{b_2 > -b_1 - 1} \tag{4}$$

$$\boxed{b_2 > b_1 - 1} \tag{5}$$

Fourier Transforms

Discrete Time Fourier Transform (DTFT)

- From discrete time to continuous frequency
- Equation 3.10 in the textbook.
- On the 2014 formula sheet

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\theta n} \tag{6}$$

Inverse Discrete Time Fourier Transform

- From continuous frequency to discrete time .
- Equation 3.14 in the textbook.
- On the 2014 formula sheet

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{j\theta n} d\theta \tag{7}$$

Continuous Time Fourier Transform (CTFT)

- From continuous time to continuous frequency.
- On the 2014 formula sheet

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Inverse Continuous Time Fourier Transform (ICTFT)

- From continuous frequency to continuous time.
- On the 2014 formula sheet

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \tag{8}$$

Discrete Fourier Transform (DFT)

- From discrete time to discrete frequency
- This is similar to the IDTFT (equation 7), but we've but we've subbed in $\theta = \frac{2\pi k}{N}$.
- This equations samples $X(z)$ at N equally spaced points around the unit circle (starting from $z = 1$).

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-jn\frac{2\pi k}{N}} \text{ for } k = 0, 1, 2, 3, \ldots (N-1)$$

Inverse Discrete Fourier Transform

- From discrete frequency back to discrete time
- This is similar to the IDTFT (equation 7), but we've but we've subbed in $\theta = \frac{2\pi k}{N}$.

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{jn\frac{2\pi}{N}k} \text{ for } n = 0, 1, 2 \ldots (N-1)$$

Linear Phase Filters

- A filter has a "linear phase response" iff $\phi(\theta) = b - a\theta$.
  - The coefficients must be symmetric
  - $h[m+n] = \pm h[m-n]$
  - $z^{-m}H(z) = \pm z^{-m}H\left(\frac{1}{z}\right)$
  - $H(z) = \pm H\left(\frac{1}{z}\right)$
  - $\therefore$The zeros of a linear phase system must occur in reciprocal pairs

Phase Delay

$$\boxed{T_p = -\frac{\phi(\theta)}{\theta}}$$

Group Delay

$$\boxed{T_g = -\frac{d\phi(\theta)}{d\theta}}$$

Linear Phase Filter Types

You are not expected to remember all of these.

The sequence has even symmetry iff there are an even number of zeros at $z = 1$.

The sequence has even length iff the number of zeros at $z = 1$ plus the number at $z = -1$ is odd.

Type 1 even number of zeros at $z = 1$ and an even number at $z = -1$.
Sequence has even symmetry and odd length.

Type 2 even number of zeros at $z = 1$ and an odd number at $z = -1$.
Sequence has even symmetry and even length.

Type 3 odd number of zeros at $z = 1$ and an odd number at $z = -1$.
Sequence has odd symmetry and odd length.

Type 4 odd number of zeros at $z = 1$ and an even number at $z = -1$.
Sequence has odd symmetry and even length.

Min, Max, Mixed Phase

Minimum phase filter A filter such that $\phi(\pi) - \phi(0) = 0$. It also happens to have the smallest $\phi(\theta)$ for all θ. All the zeros are inside the unit circle.

Maximum phase filter A filter such that $\phi(\pi) - \phi(0) = -k\pi$ (where k is the number of zeros). It also happens to have the largest $\phi(\theta)$ for all θ. All the zeros are outside the unit circle.

Mixed phase filter A filter such that $0 \geq \phi(\pi) - \phi(0) \geq -k\pi$. It also happens to have the. Some zeros are inside the unit circle, some are outside.

To move zeros from/to the inside of the unit circle, to/from the outside, take their reciprocals. The magnitude response is unaffected by this change.

All the complex zeros must occur in conjugate pairs.

Oscillators

To oscillate at frequency θ_0, place poles at $p = e^{\pm j\theta_0}$. The transfer function[3] is therefore:

$$\begin{aligned} Y(\theta) &= \frac{1}{(1 - e^{j\theta_0} z^{-1})(1 - e^{-j\theta_0} z^{-1})} \\ &= \frac{1}{1 - 2\cos(\theta_0) z^{-1} + z^{-2}} \end{aligned}$$

So the difference equation is:

$$\boxed{y[n] = 2\cos(\theta_0)\, y[n-1] - y[n-2]}$$

Maths

When doing some proofs, you may need to remember[4] "Dirichlet Kernels".

$$\sum_{k=-n}^{n} e^{ik\theta} = 1 + 2\sum_{k=1}^{n} \cos(k\theta) = \frac{\sin\left(n + \frac{\theta}{2}\right)}{\sin(\frac{\theta}{2})}$$

Analogue Filters

Butterworth

Maximally flat in passband

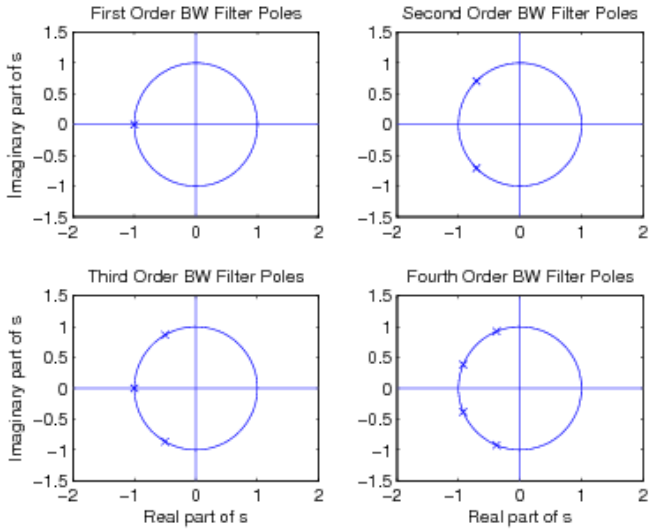
- This amplitude response formula is in the formula sheet.

$$|H_n(j\omega)| = \frac{1}{1 + \left(\frac{\omega}{\omega_{3dB}}\right)^{2n}}$$

[3] Even though there isn't an input.

[4] These were never shown in lectures, but they pop up in some of the tute questions.

- For an order n normalised low pass Butterworth filter, the poles are what you get if you spread $2n$ poles evenly around the unit circle in the s plane, and then ignore the ones in the right half s plane. For odd n there is a pole at $s = -1$. For even n there are poles either side of $s = -1$.



- The amplitude bode plot of a low pass Butterworth filter is $0dB$ from DC to $\omega = \omega_c$, then sloping downwards at $-n \times 20dB/\text{decade}$

Other Types

Chebyshev Type I Maximal ripple in passband

Chebyshev Type II Maximal ripple in stopband

Elliptic Minimum transition width, ripple in passband and stopband

Bessel Maximally constant group delay

Analogue to Analogue Conversions

Low pass unity cut off to low pass

$$s \to \frac{s}{\omega_c}$$

Low pass unity cut off to high pass

$$s \to \frac{\omega_c}{s}$$

Low pass unity cut off to band pass

- This formula will hopefully be given to us.

$$s \to \frac{s^2 + \omega_L \omega_U}{s\left(\omega_u - \omega_L\right)}$$

Low pass unity cut off to band stop

- This formula will hopefully be given to us.
- This is the inverse of the band pass transformation.

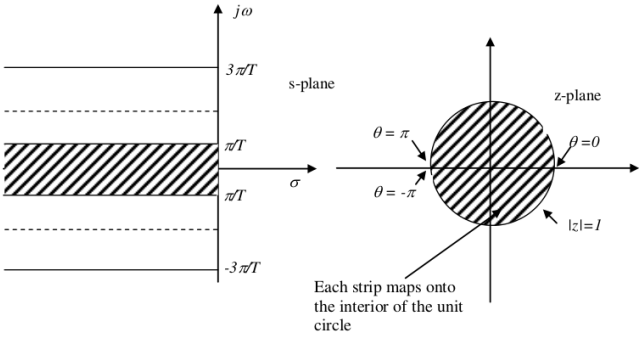
$$s \to \frac{s\left(\omega_u - \omega_L\right)}{s^2 + \omega_L \omega_U}$$

Impulse Invariant Method

- This formula is on the formula sheet.

$$\frac{1}{s+a} \to \frac{1}{1 - e^{-bT} z^{-1}}$$

- This works for $b \in \mathbb{C}$**.** (The complex parts tend to cancel out.)
- Some aliasing occurs. This is minimized by using a high sampling frequency.
- Only $\{s : \Re(s) \leq 0\} \cap \left\{s : |\Im(s)| \leq \frac{\pi}{T}\right\}$ is mapped into the unit circle in the z plane.

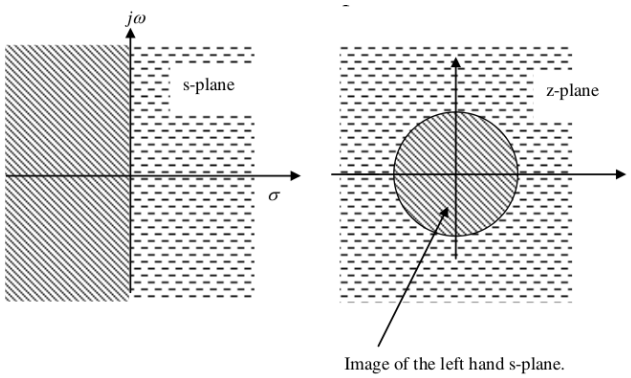


Bilinear Transformation

- This formula is on the formula sheet.

$$s = \frac{2}{T} \cdot \frac{1 - z^{-1}}{1 + z^{-1}}$$

- When solving problems with this, don't sub in $\frac{2}{T}$ into the calculator. Leave it as is and it will cancel out later.



- No aliasing occurs, but this method requires prewarping.
  - To prewarp a desired cutoff frequency θ_c to get an analogue cut off frequency:

$$\boxed{\omega_c' = \frac{2}{T}\tan\left(\frac{\theta_c}{2}\right) = \frac{2}{T}\tan\left(\frac{2\pi f_c}{2f_s}\right)}$$

  - Prewarping always increases the analogue cutoff frequency.

Proofs

You should be able to reproduce these proofs in the exam.

Signal To Quantization Noise Error (SQNR)

Question

Prove that the SQNR for a signal with RMS value σ_{sig} when fed into an ADC with B bits and range $\pm A$ is given by:

$$\text{SQNR} = 6.02B + 4.77 - 20\log\left(\frac{A}{\sigma_{sig}}\right)$$

Solution

$$\text{SQNR} = \frac{P_{in}}{P_N} \tag{9}$$

$$P_{in} = \sigma_{sig}^2 \tag{10}$$

From equation 1 we get:

$$\begin{aligned} P_N &= \frac{(\Delta V)^2}{12} \\ &= \frac{\left((2A) \times 2^{-B}\right)^2}{12} \\ &= \frac{A^2 2^{-2B}}{3} \end{aligned} \tag{11}$$

Subbing equations 10 and 11 into 9 we get:

$$\begin{aligned} \text{SQNR} &= \frac{\sigma_{sig}^2}{\frac{A^2 2^{-2B}}{3}} \\ &= 3 \times 2^{2B} \times \left(\frac{\sigma_{sig}}{A}\right)^2 \end{aligned}$$

We use $10\log_{10}$ instead of $20\log_{10}$ because equation 9 deals with power.

$$\begin{aligned} \text{SQNR}_{dB} &= 10\log\left(3 \times 2^{2B} \times \left(\frac{\sigma_{sig}}{A}\right)^2\right) \\ &= 6.02B + 4.77 - 20\log\left(\frac{A}{\sigma_{sig}}\right) \end{aligned}$$

Convolution

Question

Prove that convolution in the time domain equals multiplication in the frequency domain. i.e.

$$y[n] = x[n] * h[n] \Longleftrightarrow Y(\theta) = X(\theta) \cdot H(\theta)$$

Where the formula for convolution is:

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]\, y[n-k]$$

Solution

$$
\begin{aligned}
y[n] &= x[n] * h[n] \\
&= \sum_{k=-\infty}^{\infty} x[k]\, y[n-k] \\
Y(\theta) &= \sum_{n=-\infty}^{\infty} y[n]\, e^{-jn\theta} \\
&= \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x[k]\, y[n-k] \right) e^{-jn\theta} \\
&= \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x[k]\, y[n-k]\, e^{-jn\theta} \\
&= \sum_{k=-\infty}^{\infty} x[k] \sum_{n=-\infty}^{\infty} y[n-k]\, e^{-jn\theta} \\
&= \sum_{k=-\infty}^{\infty} x[k]\, e^{-jk\theta} \sum_{n=-\infty}^{\infty} y[n-k]\, e^{-j(n-k)\theta} \\
&= \sum_{k=-\infty}^{\infty} x[k]\, e^{-jk\theta} \sum_{n=-\infty}^{\infty} y[n]\, e^{-jn\theta} \\
&= \left(\sum_{k=-\infty}^{\infty} x[k]\, e^{-jk\theta} \right) \cdot \left(\sum_{n=-\infty}^{\infty} y[n]\, e^{-jn\theta} \right) \\
&= X(\theta) \cdot H(\theta)
\end{aligned}
$$

Stability Triangle

Question

Prove that any second order system with transfer function denominator $z^2 + b_1 z + b_2$ is stable if and only if equations 3, 4 and 5 hold.

Solutions

To prove stability we must find the conditions to place on b_1 and b_2 such that the poles lie inside the unit circle. Applying the quadratic formula to the denominator we get:

$$\begin{aligned}\left|\frac{-b_1 \pm \sqrt{b_1^2 - 4b_2}}{2}\right| &< 1 \\ \left|b_1 \pm \sqrt{b_1^2 - 4b_2}\right| &< 2\end{aligned}$$

If the roots are real :

$$b_1^2 - 4b_2 \geq 0 \tag{12}$$

$$-2 < b_1 - \sqrt{b_1^2 - 4b_2} \leq b_1 + \sqrt{b_1^2 - 4b_2} < 2$$

So we have:

$$\begin{aligned}\sqrt{b_1^2 - 4b_2} &< b_1 + 2 \\ b_1^2 - 4b_2 &< b_1^2 + 4b_1 + 4 \\ b_2 &> -b_1 - 1 \end{aligned} \tag{13}$$

And

$$\begin{aligned}\sqrt{b_1^2 - 4b_2} &< 2 - b_1 \\ b_1^2 - 4b_2 &< b_1^2 - 4b_1 + 4 \\ b_2 &> b_1 - 1 \end{aligned} \tag{14}$$

If the roots are complex:

$$b_1^2 - 4b_2 < 0 \tag{15}$$

$$\begin{aligned}b_1^2 + \left(4b_2 - b_1^2\right) &< 4 \\ b_2 &< 1 \end{aligned} \tag{16}$$

So now we have the union of (the intersection of equations 12, 13 and 14) and (the intersection of equations 15 and 16), which simplifies to

$$b_2 < 1$$

$$b_2 > -b_1 - 1$$

$$b_2 > b_1 - 1$$

These can be seen in figure 3.

Analogue Digital Equivalence

In past exams students have been asked to reproduce this proof. Doing so requires rote learning the 2 magic substitutions.

Question

Prove that:

$$X(\theta) = \frac{1}{T}\Sigma_{k=-\infty}^{\infty} X\left(\omega + k\frac{2\pi}{T}\right)$$

Solution

Using equation 8:

$$x(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} X(\omega)\, e^{j\omega t} d\omega$$

Subbing in $t = nT$

$$x[n] = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} X(\omega)\, e^{j\omega nT} d\omega$$

Rearranging with some magic:

$$x[n] = \frac{1}{2\pi}\Sigma_{k=-\infty}^{\infty}\int_{\omega=k\omega_s-\frac{\omega_s}{2}}^{k\omega_s+\frac{\omega_s}{2}} X(\omega)\, e^{j\omega nT} d\omega$$

This is true of any value of ω_s. We're going to pick $\omega_s = 2\pi f_s = \frac{2\pi}{T}$
Magic substitution 1: Let $\widehat{\omega} = \omega - k\omega_s$, $(d\widehat{\omega} = d\omega)$

$$\begin{aligned}
x[n] &= \frac{1}{2\pi}\Sigma_{k=-\infty}^{\infty}\int_{\widehat{\omega}=-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} X(\widehat{\omega}+k\omega_s)\, e^{j(\widehat{\omega}+k\omega_s)nT} d\widehat{\omega} \\
&= \frac{1}{2\pi}\Sigma_{k=-\infty}^{\infty}\int_{\widehat{\omega}=-\frac{\pi}{T}}^{\frac{\pi}{T}} X\left(\widehat{\omega}+k\frac{\pi}{T}\right) e^{j\widehat{\omega}nT} e^{j\omega_s nT} d\widehat{\omega} \\
&= \frac{1}{2\pi}\Sigma_{k=-\infty}^{\infty}\int_{\widehat{\omega}=-\frac{\pi}{T}}^{\frac{\pi}{T}} X\left(\widehat{\omega}+k\frac{\pi}{T}\right) e^{j\widehat{\omega}nT} e^{j\frac{\pi}{T}nT} d\widehat{\omega} \\
&= \frac{1}{2\pi}\Sigma_{k=-\infty}^{\infty}\int_{\widehat{\omega}=-\frac{\pi}{T}}^{\frac{\pi}{T}} X\left(\widehat{\omega}+k\frac{\pi}{T}\right) e^{j\widehat{\omega}nT} \times 1 \times d\widehat{\omega}
\end{aligned}$$

Magic substitution 2: $\theta = \widehat{\omega}T$ $(d\theta = Td\widehat{\omega})$

$$x[n] = \frac{1}{2\pi}\Sigma_{k=-\infty}^{\infty}\int_{\theta=-\pi}^{\pi} X\left(\frac{\theta}{T}+k\frac{\pi}{T}\right) e^{jn\theta}\frac{d\theta}{T}$$

Remembering $\theta = 2\pi\frac{f}{f_s}$, so $\frac{\theta}{T} = \omega$

$$x[n] = \frac{1}{2\pi}\Sigma_{k=-\infty}^{\infty}\int_{\theta=-\pi}^{\pi} X\left(\omega + k\frac{\pi}{T}\right)e^{jn\theta}\frac{d\theta}{T}$$

Then remembering equation 7, which is reproduced here for clarity:

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)\, e^{j\theta n} d\theta$$

we get

$$\begin{aligned}
\frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)\, e^{j\theta n} d\theta &= \frac{1}{2\pi}\int_{\theta=-\pi}^{\pi} \Sigma_{k=-\infty}^{\infty} X\left(\omega + k\frac{\pi}{T}\right)e^{jn\theta}\frac{d\theta}{T} \\
\int_{-\pi}^{\pi} X(\theta)\, e^{j\theta n} d\theta &= \int_{\theta=-\pi}^{\pi} \Sigma_{k=-\infty}^{\infty} X\left(\omega + k\frac{\pi}{T}\right)e^{jn\theta}\frac{d\theta}{T}
\end{aligned}$$

Now try not to cringe[5] as we simply cancel out the integrals:

$$X(\theta)\frac{1}{T}\Sigma_{k=-\infty}^{\infty} X\left(\omega + k\frac{\pi}{T}\right)$$

[5] Yeah this is a really messy proof. Any maths lecturer would mark you down for this so hard. But this is what Ambi does.